

Yuxuan Zheng 郑宇轩

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Passionate Software Engineer with expertise in parallel computing and algorithms design. Deep understanding in OS, compilers, and distributed systems. Committed to building high-performance, reliable, and elegant solutions.

Education

M.S. in CS @ New York University

01/2024 – Exp 12/2025

- GPA: 3.8/4.0

B.E. in ECE @ Shanghai Jiao Tong University

09/2019 – 08/2023

- Core GPA: 3.7/4.0

Technical Skills

- **Programming languages:** C++, C, Go, OCaml, Python, Java, SML, Scala, Shell, Elm, TypeScript.
- **Tools & DevOps:** Git, Docker, Kubernetes, Nginx, Cloud (AWS, GCP), Hadoop, Spark, Kafka.
- **Databases:** MySQL, MariaDB, PostgreSQL, SQLite, MongoDB, Redis, Hive.

Projects

Parallel implementation of hash visualization algorithm

- Implemented a **work-efficient** parallel version of OpenSSH's hash visualization algorithm using **OpenMP**.
- Achieved **embarrassingly parallel pixel rendering**, improving performance linearly on multicore CPUs.
- Applied **load-balancing** techniques to ensure scalability across different input sizes.

Infinite world – The random map generation

- Implemented algorithm based on expression trees that generates Minecraft-like infinite maps using **procedural content generation** (PCG).
- Parallelized the algorithm using **Rake-and-Compress** algorithm to reduce the span to $\mathcal{O}(\log n)$, where n is the depth of the expression tree, hence accelerated the procedural generation speed.
- Used **Raylib** to achieve a Minecraft-like game interface.

Compiler construction

- Built an end-to-end compiler in **OCaml** that transforms ML-ish source code into optimized **RISC-V** assembly.
- Implemented robust **type checking** for compile-time error detection.
- Used **Control Graph Analysis** for optimizations including constant propagation and dead-code elimination.

Work Experience

SuXiang Automobile Tech Ltd. (Startup)

Shanghai, China

Full Stack Dev Intern

08/2023 – 09/2024

- Developed a high-performance **single-page web app** using **Elm** (frontend), **Go** with **Gin** (backend), and **MariaDB**.
- Designed and deployed self-hosted infrastructure (mail server, Mattermost, Nextcloud) via **Docker**, enabling secure team collaboration.
- Automated **CI/CD** pipeline using Drone+Gitea, improving development and deployment efficiency.

Samoyed Cloud Tech, Ltd.

Shanghai, China

Big Data Infrastructure Intern

06/2022 – 02/2023

- Processed large-scale financial datasets on **HDFS** for distributed storage and computation.
- Developed **Spark**-based ETL pipelines to clean, transform, and aggregate financial data for AI/ML training.
- Integrated **Kafka** with **Spark Structured Streaming** for real-time data analysis.